2 The polynomial $\mathrm{f}(x)$ is defined by

$$\mathrm{f}(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_2x^2 + a_1x + a_0\,,$$

where $n \geqslant 2$ and the coefficients $a_0, \ldots, a_{n-1}$ are integers, with $a_0 \neq 0$. Suppose that the equation $\mathrm{f}(x) = 0$ has a rational root p/q, where p and q are integers with no common factor greater than 1, and $q > 0$. By considering $q^{n-1}\mathrm{f}(p/q)$, find the value of q and deduce that any rational root of the equation $\mathrm{f}(x) = 0$ must be an integer.

(i) Show that the nth root of 2 is irrational for $n \geqslant 2$.

(ii) Show that the cubic equation

$$x^3 - x + 1 = 0$$

has no rational roots.

(iii) Show that the polynomial equation

$$x^n - 5x + 7 = 0$$

has no rational roots for $n \geqslant 2$.

Suppose that

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_2x^2 + a_1x + a_0,$$

where $n \geq 2$, all the coefficients are integers, and $a_0 \neq 0$. Let p/q be a rational root in lowest terms (meaning p and q are integers and have no common factors greater than 1), with $q > 0$.

Since $f(p/q) = 0$, multiplying by q^{n-1} gives

$$0 = q^{n-1}f\left(\frac{p}{q}\right) = \frac{p^n}{q} + \sum_{j=0}^{n-1} a_j\, p^j q^{n-1-j}.$$

Every term except p^n/q is an integer. Therefore p^n/q is also an integer, so q divides p^n.

If $q > 1$, it has a prime divisor r. Then $r \mid p^n$, and hence $r \mid p$. This contradicts the fact that p and q have no common factor greater than 1. Thus

$$\boxed{q = 1.}$$

Consequently, **every rational root of a monic polynomial with integer coefficients must be an integer.**

(i)

Let $\alpha = \sqrt[n]{2}$, where $n \geq 2$. Then $\alpha > 0$ and

$$\alpha^n - 2 = 0.$$

Apply the preliminary result to $F(x) = x^n - 2$. If α were rational, it would be an integer.

However,

$$F(1) = -1 < 0, \qquad F(2) = 2^n - 2 \geq 2 > 0.$$

Since $F'(x) = nx^{n-1} \geq 0$, for x > 0.

x^n is strictly increasing for $x > 0$, its unique positive solution to $x^n = 2$ satisfies $1 < \alpha < 2$. There is no integer in this interval. Therefore

$$\boxed{\sqrt[n]{2} \text{ is irrational for every integer } n \geq 2.}$$

(ii)

Let $g(x) = x^3 - x + 1$. By the preliminary result, any rational root of $g(x) = 0$ must be an integer.

Differentiating,

$$g'(x) = 3x^2 - 1 = 0 \quad \Leftrightarrow \quad x = \pm\frac{1}{\sqrt{3}}.$$

Thus g is strictly increasing on $\left(-\infty, -1/\sqrt{3}\right)$, strictly decreasing on $\left(-1/\sqrt{3}, 1/\sqrt{3}\right)$, and strictly increasing on $\left(1/\sqrt{3}, \infty\right)$.

At the two turning points,

$$\begin{aligned} g\left(-\frac{1}{\sqrt{3}}\right) &= -\frac{1}{3\sqrt{3}} + \frac{1}{\sqrt{3}} + 1 = 1 + \frac{2}{3\sqrt{3}} > 0, \\ g\left(\frac{1}{\sqrt{3}}\right) &= \frac{1}{3\sqrt{3}} - \frac{1}{\sqrt{3}} + 1 = 1 - \frac{2}{3\sqrt{3}} > 0. \end{aligned}$$

In particular, the local minimum is positive. On the interval between the turning points, g decreases to this positive minimum, and beyond the minimum it increases. Hence

$$g(x) > 0 \qquad \text{for every } x \geq -\frac{1}{\sqrt{3}}.$$

On $\left(-\infty, -1/\sqrt{3}\right)$, g is strictly increasing, with

$$\lim_{x \to -\infty} g(x) = -\infty, \qquad g\left(-\frac{1}{\sqrt{3}}\right) > 0.$$

By continuity, it has exactly one root on this interval. Therefore $g(x) = 0$ has exactly one real root in total.

Now evaluate

$$g(-2) = -8 + 2 + 1 = -5 < 0, \qquad g(-1) = -1 + 1 + 1 = 1 > 0.$$

By the intermediate value theorem, the unique real root lies in $(-2, -1)$. Since this interval contains no integer, the root cannot be rational. Hence

$$\boxed{x^3 - x + 1 = 0 \text{ has no rational roots.}}$$

(iii)

Let $h_n(x) = x^n - 5x + 7$. Again, any rational root must be an integer.

We compare the graphs of $y_1 = x^n$ and $y_2 = 5x - 7$. A root of $h_n(x) = 0$ corresponds to an intersection of the two graphs. It is therefore sufficient to show that they do not intersect at any integer value of x.

For $0 \leq x < 7/5$, we have $y_1 = x^n \geq 0$ and $y_2 = 5x - 7 < 0$, so there are no intersections. This excludes the integers 0 and 1.

When $x = 2$,

$$y_1 = 2^n \geq 2^2 = 4 > 3 = y_2.$$

When $x = 3$,

$$y_1 = 3^n \geq 3^2 = 9 > 8 = y_2.$$

For $x \geq 3$, since $n \geq 2$,

$$y_1' = nx^{n-1} \geq 2 \cdot 3 = 6 > 5 = y_2'.$$

Thus $y_1 - y_2$ is strictly increasing for $x \geq 3$. As $y_1 > y_2$ at $x = 3$, we have $y_1 > y_2$ for every $x \geq 3$. Together with the checks above, this excludes every non-negative integer root.

Now consider negative integers.

Case 1: n is even. For $x < 0$, $y_1 = x^n > 0$ whereas $y_2 = 5x - 7 < 0$. Hence the graphs cannot intersect, and there are no negative roots. Therefore there are no rational roots when n is even.

Case 2: n is odd

Since $n \geq 2$, an odd n must satisfy $n \geq 3$. At $x = -1$,

$$y_1 = (-1)^n = -1 > -12 = y_2,$$

so $x = -1$ is not a root.

For $x \leq -2$, $n - 1$ is even, so

$$y_1{}' = nx^{n-1} = n|x|^{n-1} \geq 3 \cdot 2^2 = 12 > 5 = y_2{}'.$$

Hence $h_n(x) = y_1 - y_2$ is strictly increasing on $(-\infty, -2]$.

If $n = 3$, at $x = -2$,

$$y_1 = -8 > -17 = y_2, \qquad h_3(-2) = 9 > 0.$$

At $x = -3$,

$$y_1 = -27 < -22 = y_2, \qquad h_3(-3) = -5 < 0.$$

Since h_3 is strictly increasing for $x \leq -2$,

$$h_3(x) \leq h_3(-3) = -5 < 0 \qquad \text{for all } x \leq -3.$$

Thus every integer at most -3 gives a negative value, while -2 and -1 give positive values. None is a root.

If $n \geq 5$ is odd, at $x = -2$,

$$y_1 = -2^n \leq -32 < -17 = y_2,$$

so

$$h_n(-2) = 17 - 2^n \leq -15 < 0.$$

By the same monotonicity,

$$h_n(x) \leq h_n(-2) < 0 \qquad \text{for all } x \leq -2.$$

This excludes every integer at most -2, and -1 has already been excluded. Hence there are no negative integer roots for odd $n \geq 5$ either.

Combining all cases, there are no integer roots for any $n \geq 2$. Since every rational root must be an integer,

$$\boxed{x^n - 5x + 7 = 0 \text{ has no rational roots for every integer } n \geq 2.}$$